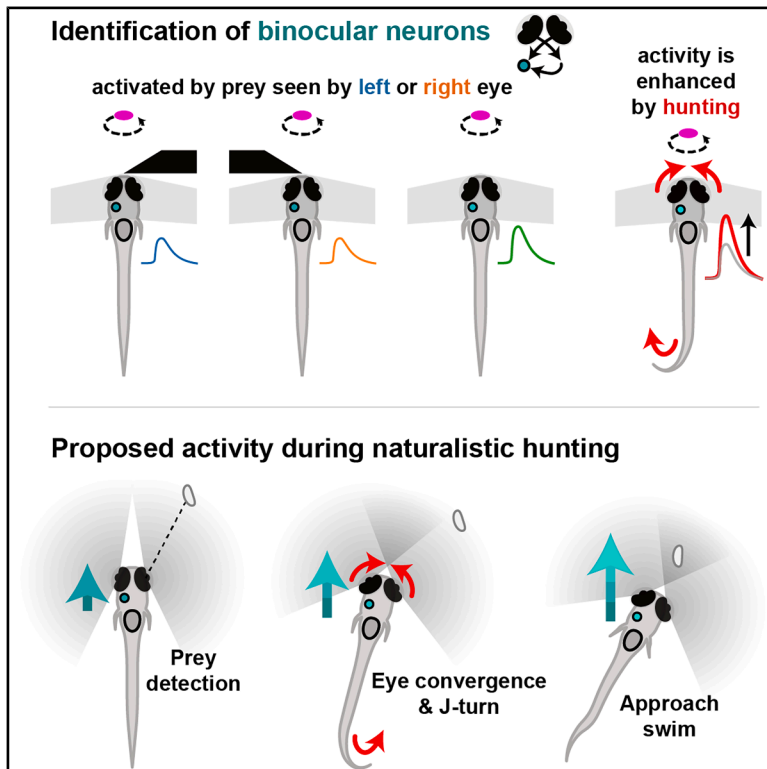


Current Biology

Binocular integration of prey stimuli in the zebrafish visual system

Graphical abstract



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In brief

Tian et al. identify binocular neurons in the zebrafish visual system that respond to prey stimuli viewed by the left or the right eye. These neurons are activated when larvae initiate hunting behavior, suggesting that they could be important for the detection and pursuit of prey objects in the binocular zone.

Highlights

- Large-scale imaging identifies a population of binocular prey-responsive neurons
- These neurons are located in the pretectum, thalamus, and nucleus isthmi
- Binocular neurons should respond to prey objects within striking distance
- Binocular neurons may receive a hunting-related corollary discharge



Article

Binocular integration of prey stimuli in the zebrafish visual system

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<https://doi.org/10.1016/j.cub.2025.05.063>

SUMMARY

Most animals with two eyes combine the inputs to achieve binocular vision, which can serve numerous functions and is particularly useful in hunting prey. However, the mechanisms by which visual information from the two eyes are combined remain largely unknown. Here, we designed a device to reversibly occlude the eyes of a head-fixed zebrafish larva, and we used large-scale volumetric two-photon imaging to identify binocular neurons that respond to prey stimuli. We found that these binocular prey-responsive neurons (bino-PRNs) are primarily located in three areas, the pretectum, thalamus, and nucleus isthmi. We then characterized the bino-PRNs' functional properties and found that their left and right eye receptive fields are offset to varying degrees, which would correspond to objects at naturalistic hunting distances for a larva with converged eyes. We also found that bino-PRNs have a significantly greater response in hunting trials, which could be the result of an eye convergence-related corollary discharge. We then optogenetically induced prey capture eye and tail movements and found that this hunting command activates PRNs in the pretectum, thalamus, and nucleus isthmi. These findings indicate that bino-PRNs receive visual and motor input that would allow them to encode prey position in three dimensions.

INTRODUCTION

All animals with binocular vision must integrate the two eyes' signals to produce a unified percept and behavioral response, and this integration should occur in binocular neurons that receive input from both eyes. Binocular neurons in the visual cortex have been identified in non-human primates as well as mice, but it remains unclear whether the key binocular interactions first occur in the lateral geniculate nucleus (LGN) or visual cortex and how the inputs from the two eyes are combined.^{1,2} One proposed driver of the evolution of binocular vision is predation,³ and binocular vision has been proposed to play a key role in hunting in a diverse range of animals, from insects to cephalopods to mice.^{4–6} Mice have been shown to place prey objects in their binocular visual field, and ipsilateral retinal ganglion cell input is important for hunting.⁷ However, it has been difficult to identify the binocular neurons that respond to the prey and to manipulate these neurons to determine their role in hunting behavior.

Binocular vision is also thought to play a role in zebrafish hunting behavior.⁸ Zebrafish prey capture (PC) is typically

initiated by the entry of a prey object into the peripheral visual field. In response to the prey stimulus, the larva converges its eyes and turns toward the prey. These actions bring the prey from the peripheral to the central visual field,^{9–13} after which the larva may perform a forward swim to approach the prey, before executing a capture strike to consume it.^{10,12} Zebrafish larvae have laterally positioned eyes that normally have little binocular overlap, but the eye convergence at the beginning of a hunting episode creates a binocular zone.⁹ Since eye convergence is a signature of hunting,⁹ it is likely that binocular vision plays a key role in PC, although its exact role remains unclear.

There are multiple possible functions for binocular vision. First, perhaps the simplest application is to increase the prey signal by summing two visual inputs, which could help larvae detect low-contrast prey.¹⁴ A second possibility is that larvae use binocular vision for range finding or estimating the distance to prey, which could be achieved by foveating the prey with both eyes and triangulating the distance, if the extent of eye convergence is known.¹⁵ A third option is that they are doing a more sophisticated form of stereopsis and are detecting



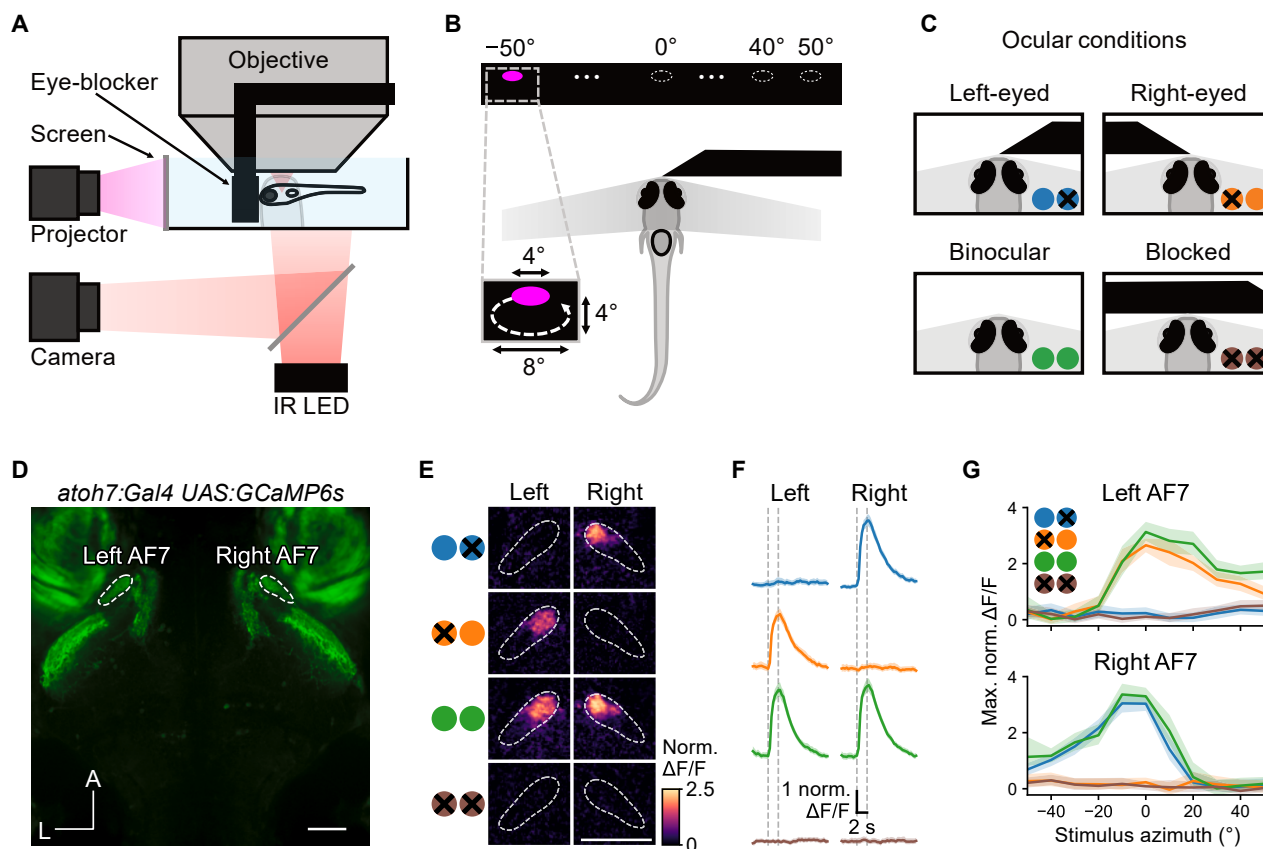


Figure 1. Reversible occlusion of one eye blocks the response to prey stimuli

(A) Schematic of the experimental setup: the eye-blocker is positioned in front of the fish to block visual input.
 (B) Bright UV circulating prey-like stimuli were presented at azimuths from -50° (left) to $+50^\circ$ (right).
 (C) Larvae were tested under 4 ocular conditions: right eye blocked, left eye blocked, neither eye blocked, and both eyes blocked.
 (D) Baseline fluorescence image showing AF7s labeled with GCaMP6s (scale bar, 50 μm).
 (E) Activation of AF7 in left eye, right eye, binocular, and fully blocked conditions in one example larva (scale bar, 50 μm). Pixels are color-coded by trial-averaged increase in mean $\Delta F/F$ during the stimulus interval, compared with that during the 2 s before the stimulus.
 (F) Response of the left and right AF7 to 0° prey stimulus. Vertical dashed lines indicate the stimulus interval. Solid line and shaded region represent the mean and 95% confidence interval, respectively ($n = 7$ fish).
 (G) Spatial tuning curves showing maximum activity of AF7 (within the 3-s time window after stimulus onset) in response to prey stimuli at different azimuths. Solid lines and shaded regions represent the mean and 95% confidence interval, respectively ($n = 7$ fish).
 See also Figure S1.

binocular disparity, or the difference in the retinal position of the prey between the two eyes, to perceive depth.¹⁶ The extent to which larvae use binocular vs. monocular cues for distance estimation also remains unknown.^{17–19}

One way to clarify the role of binocular vision would be to find binocular neurons and study their functional properties. So far, although binocular interactions have been identified in the visual response to whole-field motion and luminance stimuli,^{20–22} binocular prey-responsive neurons (bino-PRNs) have not been found in zebrafish.

Here, we use large-scale imaging to identify bino-PRNs in multiple visual areas of the larval zebrafish brain. We find that these neurons are tuned to objects within striking distance of a hunting larva, and they also integrate eye convergence information, which could allow them to determine the location and distance of prey objects.

RESULTS

Eye-blocker reversibly blocks input to one eye

To identify bino-PRNs, we designed a device to block one or both eyes selectively and reversibly by laterally moving its arms in front of the eyes (Figures 1A and 1B). Larvae were embedded in agarose with the tail free, and the eyes were kept in agarose to minimize large eye movements, which would change neuronal receptive fields. Although the eyes were fixed, we could observe eye convergences of a few degrees in response to small, moving UV dots, which trigger hunting behavior.¹⁷ In order to restrict the prey azimuth, we designed a circulating stimulus that moved through an $8^\circ \times 4^\circ$ elliptical track, and we presented this at azimuths from -50° (left) to $+50^\circ$ (right) of the visual field (Figure 1B). All stimuli were presented during each ocular condition: left eye exposed (left-eyed), right eye

exposed (right-eyed), both eyes exposed (binocular), and both eyes blocked (blocked) (Figure 1C). We tested the completeness of the block by recording visual responses in arborization field 7 (AF7), a retinorecipient region that responds to prey stimuli,²³ and found that blocking one eye completely blocked the input to AF7 in the contralateral hemisphere (Figures 1D and 1E). This pattern was consistent across seven fish (Figure 1F) and all stimulus azimuths (Figure 1G).

We found that blocking one eye affected behavior; in the monocular condition, larvae would respond to prey up to 10° on the blocked side but not to more lateral stimuli, indicating that the binocular field in our eyes-fixed condition is $\pm 10^\circ$ (Figure S1A). For central prey stimuli, the occlusion also caused larvae to bias the initial hunting tail beat to the exposed eye side (Figure S1B), and bout asymmetry was higher in monocular conditions (Figure S1C), indicating that the blocker affected hunting behavior.

One population of PRNs is binocular

We next used two-photon imaging to identify PRNs in binocular and monocular conditions. We used an electrically tunable lens to image a $500 \times 500 \times 100 \mu\text{m}$ volume, which included the optic tectum (OT), pretectum (Pt), tegmentum (Teg), nucleus isthmi (NI), and thalamus (Thal) (Figure 2A). All these areas have previously been shown to respond to prey stimuli.^{24–30} For larvae expressing nuclear-localized GCaMP6s pan-neuronally (*Tg(elavl3:Hsa.H2B-GCaMP6s)/jif5Tg*, aka *elavl3*),³¹ we presented the same prey stimuli as in Figure 1 while recording calcium responses in 9 planes of the brain at 3 volumes per second, and mapped the positions of the cell bodies onto the mapZebbrain atlas³² (Figure S2A). We moved the eye-blocker to different positions with respect to the larva to record the responses in the left-eyed, right-eyed, binocular, and fully blocked conditions (Figures 2B, 2C, and S2B). We also recorded eye and tail movements and used eye convergence as a marker for hunting. Although the eyes were in agarose, we could still observe convergences of a few degrees, and we used these small convergences (4.5° binocularly, on average) to classify hunting behavior (Figure 2C; Video S1; STAR Methods). These convergence events occurred almost exclusively during prey presentation (Figure S2C). To identify PRNs, we modeled the activity of each neuron as a weighted sum of stimulus regressors (Figure S2D). We found that neurons with $R^2 \geq 0.3$ were well correlated with prey stimuli (Figure S2E) and that the fully blocked condition resulted in virtually no PRNs above this threshold (Figure S2F). Therefore, we defined neurons with $R^2 \geq 0.3$ as prey responsive. We used spatial *p* value analysis³³ to select neurons that were co-localized across fish. We found that while the majority of the responsive neurons were in the hemisphere contralateral to the unblocked eye, we also observed a smaller population of PRNs in the hemisphere ipsilateral to the unblocked eye, and we termed these neurons “ipsi-neurons” (Figure 2E).

We then asked how the population of PRNs responded across different ocular conditions (Figure 2E). As expected, a large class of PRNs only responded to contralateral eye input (mono-contra; Figure 2F). We also found that a small population within the ipsi-neurons was unresponsive to contralateral eye input (Figure S2G). Finally, one population of ipsi-neurons also

responded to input from the contralateral eye (binocular; Figure 2G). We identified the anatomical locations of the bino-PRNs, and we found that they were predominantly mapped to three areas: the Pt, Thal, and NI (Figure 2G, bar plot). The number and locations of bino-PRNs varied somewhat between individual larvae, but out of the 15 larvae, 11 had bino-PRNs in the Pt and Thal, and 6 had bino-PRNs in the NI (Figure S2H).

To confirm that the bino-PRNs are not an artifact of incomplete eye occlusion, we also performed lensectomies to remove one lens of the larvae, which renders larvae unable to respond to prey¹² (Figure S3A). The PRN patterns of sham, unilateral, and bilateral delensed fish resembled those produced by the eye-blocker, and ipsi-neurons were found in the Pt, Thal, and NI of unilaterally delensed larvae (Figures S3B and S3C).

Bino-PRNs respond to prey-like stimuli and not to whole-field motion or looming stimuli

Next, we investigated whether bino-PRNs are activated by other behaviorally relevant visual stimuli. As zebrafish use red cones to detect global motion,^{34,35} we introduced a red background to our visual stimuli. We collected a second imaging dataset, in which larvae were presented with circulating prey dots, as in Figure 2, and then with a battery of eight additional stimuli (darkening/brightening, sweeping UV dots, horizontally moving gratings, and looming disks) in the binocular condition (Figure 3A).

We identified bino- and mono-PRNs as in Figure 2 (Figure 3B). Using affinity propagation clustering followed by hierarchical clustering,²⁸ neurons were grouped into functional types (Figure 3C). The vast majority of bino-PRNs were assigned to cluster 2 (Figures 3D and 3E), which responded to both leftward and rightward moving dots (Figure 3F). In summary, bino-PRNs respond to prey-like stimuli with circular motion or sweeping in both horizontal directions, rather than to dark moving gratings, looming stimuli, or changes in luminance. However, we did not test other non-prey UV stimuli, so bino-PRNs could be UV specific and not prey specific.

Bino-PRN receptive fields

To investigate the potential visual functions of the bino-PRNs, we next mapped their left eye and right eye receptive fields. Since we presented the prey stimulus at 12 azimuths, for each neuron, we could make left eye and right eye tuning curves for horizontal prey positions (Figure 4A), where each azimuth value represents the center of the 8° wide elliptical path of the prey stimulus (Figure 1B). We found that for some neurons, the two eyes' tuning curves are both close to the midline and are separated only by about 10° (Figure 4A, example neuron 1), while in others, the peaks of the tuning curves were more lateral (Figure 4A, example neuron 2). For these analyses, the peak azimuth could be affected by eye convergence; however, the 2° convergence that we see in hunting trials would cause a shift of about 2° on the screen, which is relatively small, compared with the 12° -wide area covered by prey stimuli. We plotted peak separations for all our bino-PRNs, and we found that there is a broad range of left to right eye peak separation distances, from 10° to 50° (Figures 4B and S4A–S4C). This range of peak separations could reflect neurons that are activated by objects at different distances from the hunting larva. In our imaging experiments, the

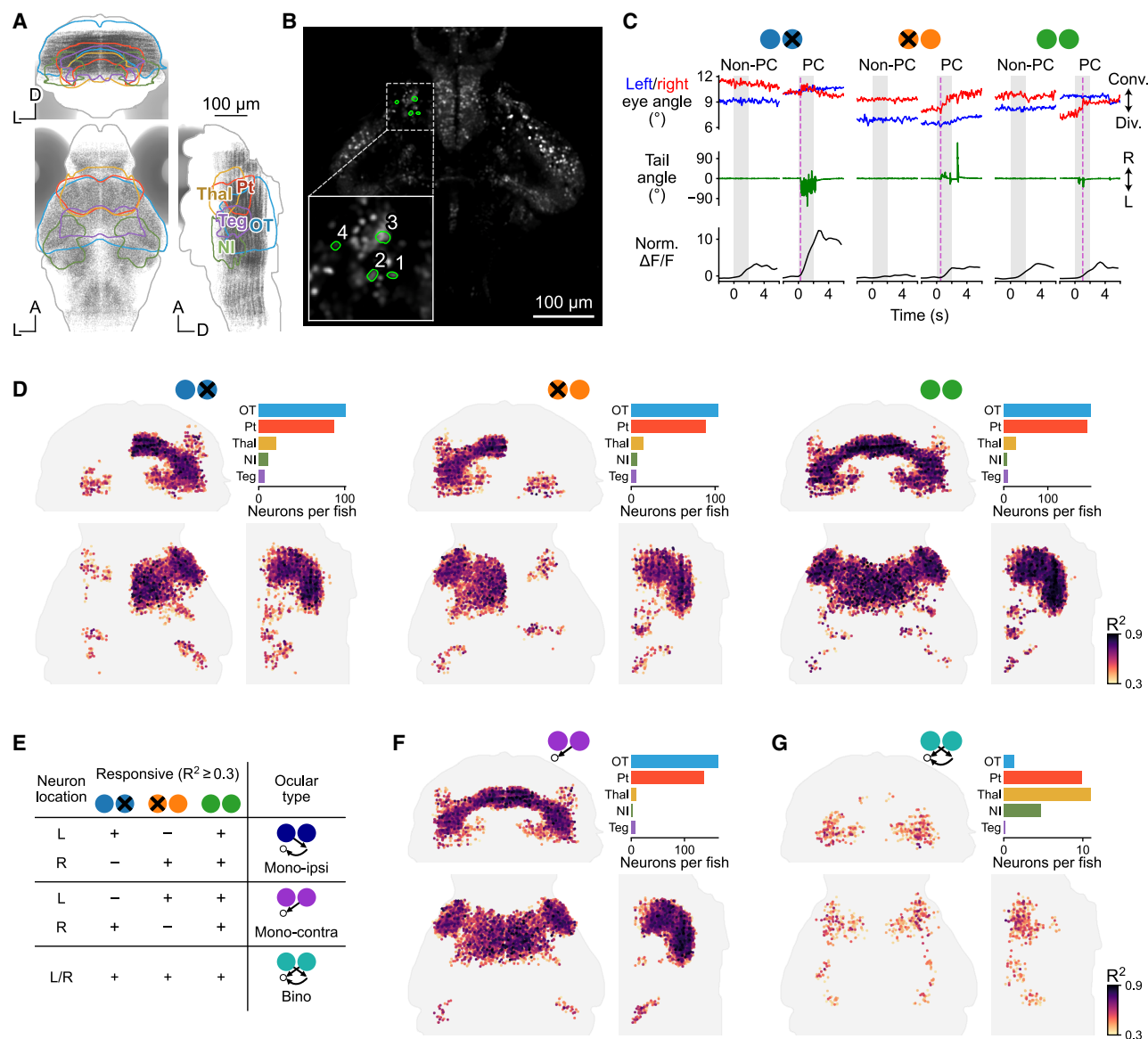


Figure 2. Volumetric two-photon imaging in different ocular conditions identifies bino-PRNs

(A) Map of the imaging volume. Each gray dot represents one neuron. Coronal, horizontal, and sagittal views (all anatomical maps follow the same layout unless otherwise stated).

(B) Maximum activity (within the 3-s time window after stimulus onset) of each neuron in response to stimuli at different azimuths in the three ocular conditions. Solid lines and shaded regions represent mean and ± 1 SD, respectively ($n = 8$ trials).

(C) Eye and tail movements triggered by prey stimuli and normalized $\Delta F/F$ responses of example neuron no. 1. Magenta dashed lines indicate hunting onsets (defined by eye convergence).

(D) Maps of PRNs in different ocular conditions from coronal, horizontal, and sagittal views ($n = 15$ fish). Neurons with prey-related activity were identified with regression analysis for each ocular condition, using an R^2 threshold of 0.3. Color code represents R^2 value. Bar plots show average number of neurons in selected brain regions.

(E) Schematic of the assignment of ocular type, based on responsiveness in different ocular conditions.

(F and G) Maps of the anatomical locations of monocular contra-PRNs (mono-contra) and bino-PRNs.

See also [Figures S2–S4](#) and [Video S1](#).

eyes were fixed and could only converge a few degrees, but in natural conditions, the eyes can converge further, depending on the distance to the prey.¹⁰ A neuron with a peak separation of 10° in our eyes-fixed imaging preparation (vergence angle $\sim 15^\circ$) would be optimally activated by an object 0.2 mm from the larva when the eyes are fully converged ([Figure 4C](#), blue

lines), whereas a neuron with a peak separation of 50° would be optimally activated by an object 1.29 mm away ([Figure 4C](#), violet lines). When the eyes are partially converged, these neurons would be best activated by more distant objects ([Figure 4C](#), 60° vergence), with a range similar to the maximum detectable prey distance (calculation: [Figure S4B](#)).

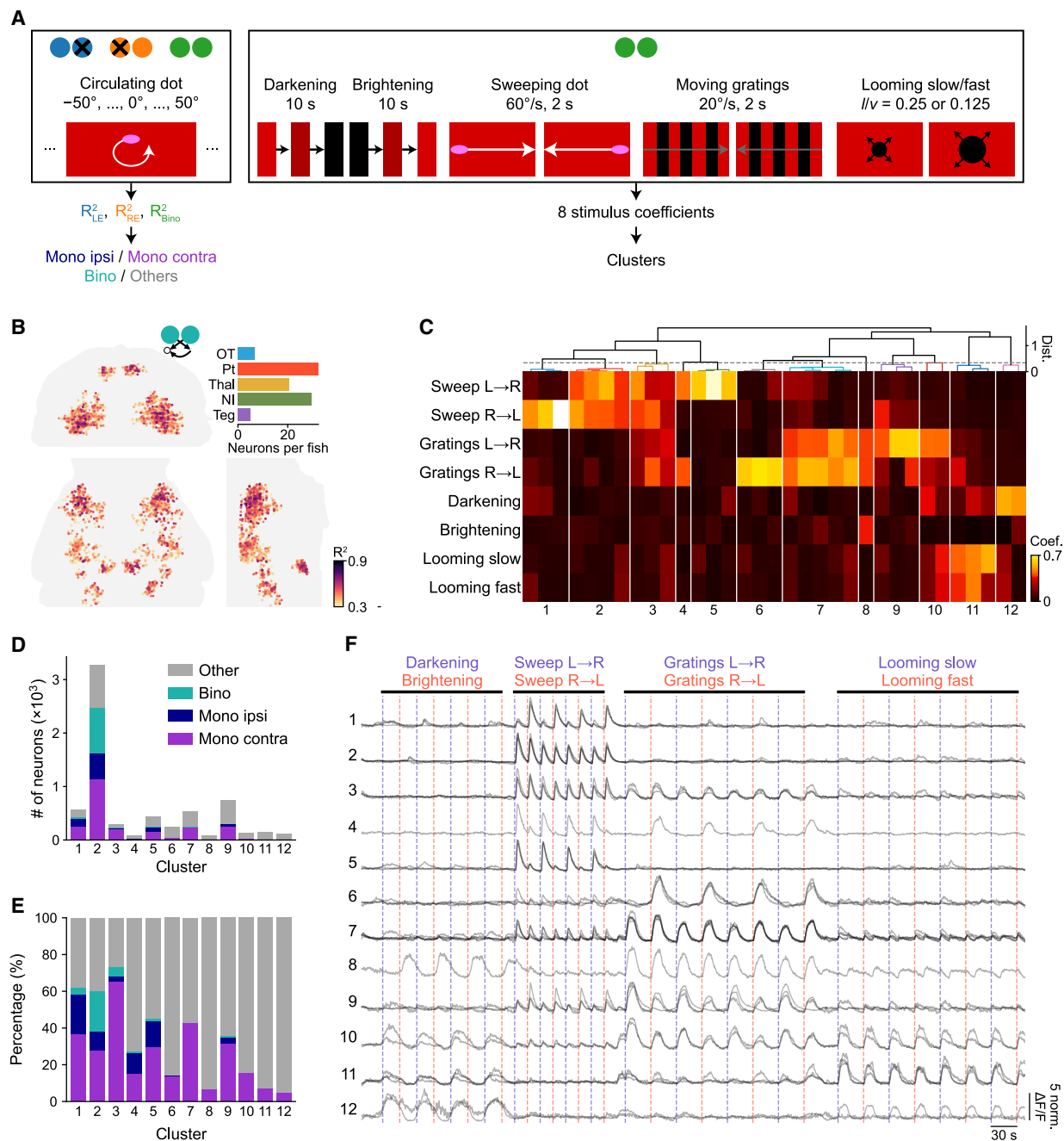


Figure 3. Bino-PRNs respond specifically to prey stimuli

(A) Schematic of the stimulus protocol. Circulating UV prey-like stimuli were presented on a red background in different ocular conditions, and a battery of eight other types of stimuli were presented binocularly to assess the specificity of bino-PRNs.

(B) Bino-PRNs were found in the Pt, Thal, and NI, using the same method as in Figure 2.

(C) Hierarchical clustering revealed 12 clusters that responded primarily to prey, moving gratings, or looming stimuli.

(D and E) Number and percentage of neurons in each cluster that are bino- or mono-PRNs.

(F) Calcium traces of all exemplars (gray) from each cluster. Blue and red vertical lines represent repeats of the two different stimuli within each stimulus category. Different shades of gray represent the activity of different subclusters within each cluster shown in (C) ($n = 10$ fish).

Thus, bino-PRNs have left and right eye receptive fields that are offset over a range corresponding to naturalistic prey distances and could thus combine binocular visual information with the degree of convergence to determine prey position.

Bino-PRNs show enhanced activity during hunting trials

Bino-PRNs appeared to respond more robustly in hunting trials than non-hunting trials (Figure 2C), and to further investigate this property, we plotted the responses of bino-PRNs in each

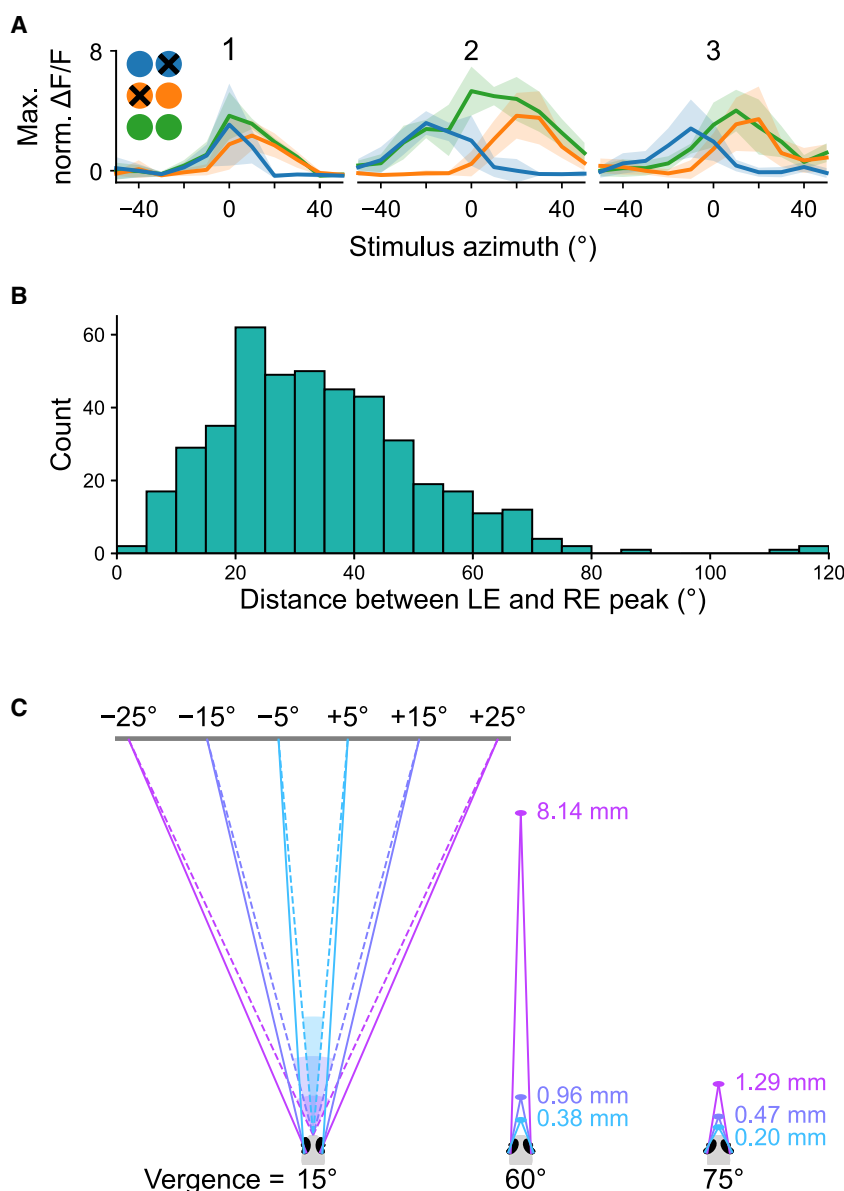


Figure 4. Bino-PRNs receive input from the two eyes that are offset in horizontal space and correspond to objects at different distances from a hunting larva

(A) Monocular and binocular tuning curves of three example bino-PRNs from one larva. Blue line represents the left eye response to stimuli presented at each azimuth, orange line represents the right eye response, and green is the binocular condition. Shading represents 1 SD.

(B) Distribution of the distances between the peaks of the left eye and right eye receptive fields for all bino-PRNs. Degree values indicate distance between the best left-eye (LE) and right-eye (RE) stimulus in screen coordinates.

(C) Schematic showing how left eye/right eye peak separations in the imaging preparation correspond to object distances at medium (60°) and maximum (75°) levels of convergence. Blue lines = 10° separation, purple lines = 30° separation, violet lines = 50° separation between right and left eye peaks. See also Figure S4.

a smaller percentage of contra-PRNs (20% in the binocular condition) show a significant enhancement (Figure 5C). This indicates that motor enhancement is not a global signal but is more robust in the bino-PRN population. We looked at the locations of these motor-modulated bino-PRNs and found that they were distributed among the Pt, Thal, and NI, the three areas with the largest number of bino-PRNs (Figure 5D). The finding that bino-PRNs are more responsive during hunting suggests that they are either generating the hunting motor command or receiving a corollary discharge when hunting is initiated.

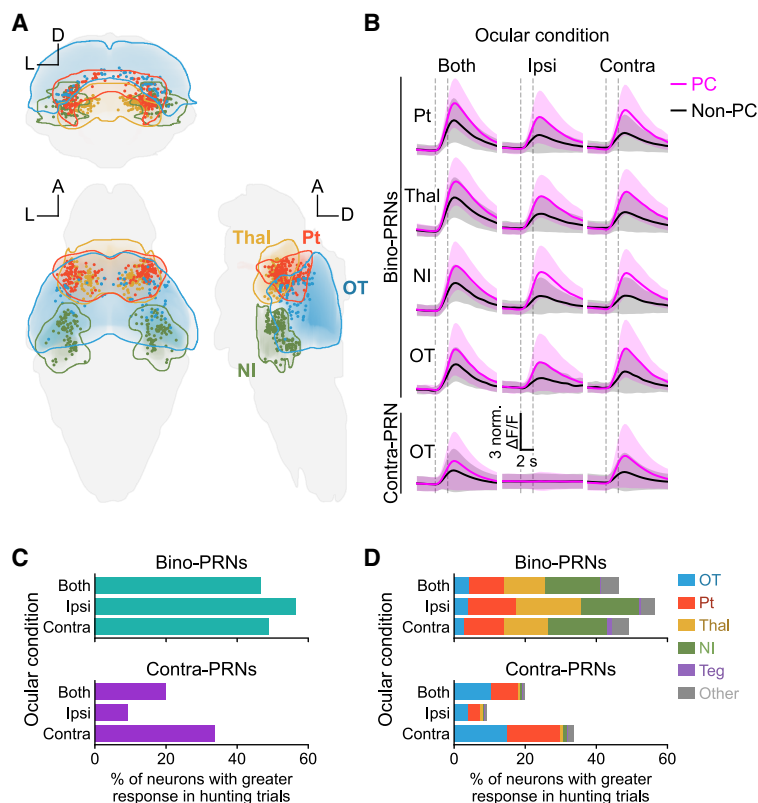
It is possible that the small convergent eye movements in hunting trials could account for the motor enhancement, due to the change in the location of the prey on the retina after convergence. To address

brain region (Figure 5A) in PC vs. non-prey capture (non-PC) trials. Indeed, the bino-PRNs in each area were on average more activated in PC trials (red lines, Figure 5B). Thus, these neurons could be binocular as a result of receiving a motor signal rather than visual input from both eyes. However, their binocularity was not entirely a result of the hunting state, as they were activated by prey stimuli seen by the ipsilateral eye even in non-hunting trials (Figure 5B, middle column, black lines). In contrast, the contra-PRNs in the OT, which constitute the majority of the PRN population, had no response to ipsilateral prey, even in hunting trials (Figure 5B, lowest row). This suggests that the bino-PRNs are receiving sensory input from both eyes.

We next looked at motor enhancement at the individual neuronal level by asking whether individual PRNs respond significantly more in hunting than non-hunting trials. We find that a large fraction of bino-PRNs are significantly more responsive during trials with eye convergence (48% in the binocular condition), whereas

this, we looked at monocular hunting trials in which the exposed eye converged less than 0.1° , and the hunting event was identified based on convergence of the blocked eye (Figure S5A). For these trials, we found that the hunting response was still greater than no hunting (Figure S5B), indicating that the motor enhancement is not due to the movement of the eye itself.

In addition to hunting modulation, another interesting bino-PRN property was their greater response in the binocular condition. Across all areas, bino-PRNs had a larger population response in the binocular than monocular condition, particularly when comparing non-hunting trials (black lines, Figure 5B). If bino-PRNs are used to determine prey distance based on the eye convergence angles and the presence of prey in the binocular visual field, we might expect them to require binocular input for full activation, as the input from one eye can indicate the presence of prey along its diagonal sight line but not the distance to prey. We analyzed binocular enhancement on a single-neuron level by calculating a



binocular enhancement index for each neuron, dividing the neuron's average of the maximal normalized $\Delta F/F$ in the binocular condition by its largest monocular response. Since binocular enhancement was more obvious in non-hunting trials, we focused on the three fish with >3 non-hunting trials to prey at 0°, the position where prey can be seen by both eyes. We found a population of bino-PRNs with a binocular enhancement index greater than 1 for non-hunting trials, (Figures S5C and S5D), whereas in contra-PRNs they were evenly distributed around 1 (Figure S5D). This binocular enhancement could be consistent with a model in which bino-PRNs combine binocular information to determine prey distance. However, due to the 1 cm distance to the screen and the fact that the eyes were fixed, our prey stimulus may not optimally stimulate both eyes, so further experiments will be required to know the full extent of binocular enhancement.

Binocular enhancement raises the question of whether there could be some neurons that respond specifically in the binocular condition (bino-specific PRNs). We thus searched for neurons with $R^2 \geq 0.3$ in the binocular condition in the same three fish. We identified 3,075 such neurons and found that 416 of them were significantly bino enhanced ($p < 0.05$ based on the fraction of the null distribution with a greater binocular enhancement index), and out of these, 50 neurons had a binocular enhancement index > 2 for prey at 0°. We then classified these neurons based on their R^2 for different ocular conditions, with the addition of a "bino-specific" category (bino $R^2 \geq 0.3$ and mono $R^2 < 0.3$), and plotted their tuning curves (Figure S5E). Most of these neurons were classified as mono-contra, as they were activated by one of the monocular conditions, but their larger binocular response at 0° lead to a high binocular enhancement index. Only

10 neurons met the R^2 criteria for bino-specific neurons (Figure S5E), suggesting that they comprise a small population of neurons in our dataset.

The *lhx9:Gal4* line labels bino-PRNs

We next sought to identify Gal4 lines labeling the bino-PRNs as a genetic marker to image and manipulate these neurons. We performed functional imaging of several lines in different ocular conditions. We first tested the *KalTA4u508* line, which was reported to express KalTA (a Gal4 variant) in the hunting command-like neurons of the accessory pretectal nucleus (APN) (Figure 6A).²⁵ When we imaged *Tg(KalTA4u508;UAS:GCaMP6s)* larvae, we found that almost all PRNs were mono-contra (Figures 6B–6D). All the pretectal PRNs in the line were mono-contra, meaning that the pretectal command-like neurons must be receiving input from only the contralateral eye.

In contrast, when we imaged *Tg(lhx9:Gal4;UAS:GCaMP6s)* larvae³⁶ (Figure 6E), we found that the majority of PRNs were binocular. The *lhx9+* bino-PRNs were located in the Pt (as well as Thal) and the anterior parts of the NI (Figures 6F–6H). These neurons were activated by prey presented to either the contralateral or ipsilateral eye (Figure 6J) and by ipsilateral prey even in the absence of hunting, similar to the bino-PRNs in the panneuronal dataset (Figure 5B). These data suggest that the *lhx9+* bino-PRNs represent a separate population that is not labeled by the *KalTA4u508* line. We can therefore use these two lines to investigate the source of motor enhancement of the bino-PRNs.

Optogenetically triggered hunting events activate *lhx9+* Pt, Thal, and NI neurons in the absence of prey

Based on the enhanced activity of bino-PRNs in hunting trials (Figure 5), we hypothesized that they could be activated as a consequence of hunting. Zebrafish larvae normally have unconverged eyes, to maximize coverage of the visual field, and only converge their eyes at the onset of a hunting episode.^{9,10,13} It may be that to properly interpret binocular visual input, some information about hunting onset or the degree of convergence has to be conveyed to visual areas. One way for this information to

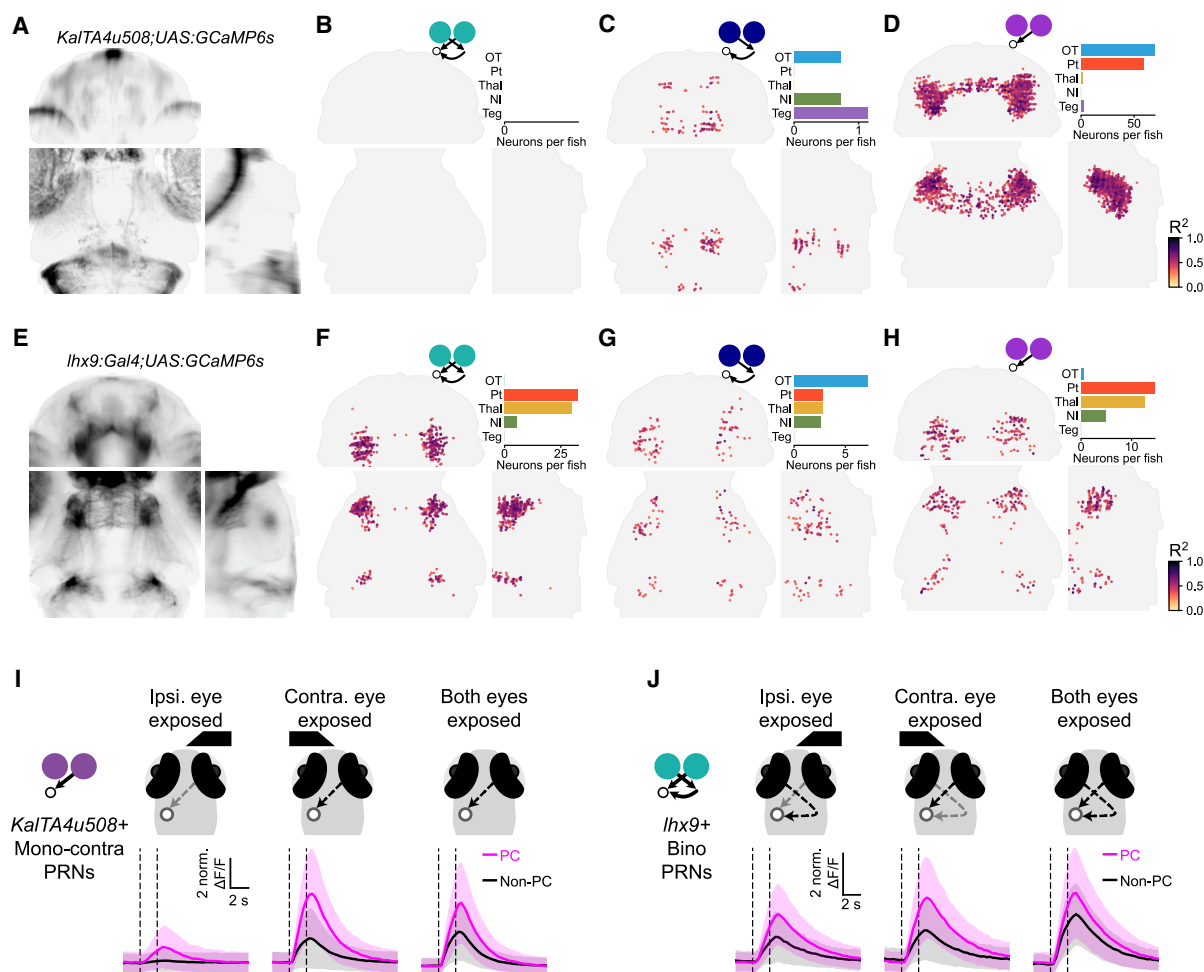


Figure 6. Identification of a transgenic line labeling bino-PRNs

(A) Expression pattern of the *KalTA4u508* line.

(B–D) Binocular, mono-ipsi, and mono-contra PRNs in the *u508* line ($n = 7$ fish).

(E) Expression pattern of the *Ihx9* line.

(F–H) Binocular, mono-ipsi, and mono-contra PRNs in the *Ihx9* line ($n = 5$ fish).

(I) Average response of the mono-contra PRNs in the *KalTA4u508* line in PC and non-PC trials. Solid lines and shaded regions represent mean and ± 1 SD, respectively.

(J) Average response of the bino-PRNs in the *Ihx9* line. Solid lines and shaded regions represent mean and ± 1 SD, respectively.

get there would be via corollary discharge or an internal signal that is the result of a motor command.³⁷ Thus, visual areas could receive a signal that is the result of the hunting motor command, which would indicate that the eyes are converged and that the animal is hunting. However, it is also possible that PRNs are more responsive during hunting trials because they are playing a role in generating the motor command.

To distinguish between these possibilities, we used holographic optogenetic activation of hunting command-like neurons to artificially induce PC (opto-hunting), and we then recorded the activity of PRNs to observe whether they receive an excitatory input in opto-hunting trials. The *KalTA4u508* line was previously shown to label neurons in the APN whose activation can evoke PC.²⁵ We hypothesized that bino-PRNs could be receiving a corollary discharge input, directly or indirectly, from the APN after hunting initiation (Figure 7A). If this was the case,

we would expect that artificial activation of hunting via optogenetic stimulation of the APN should cause bino-PRN activation. Alternatively, the bino-PRNs could be upstream of the APN (Figure 7B), in which case they would not be activated as a consequence of APN stimulation.

To activate the APN and record the neurons that were activated during opto-hunting events, we expressed GCaMP6s and channelrhodopsin (ChR2) in the *u508* and *Ihx9*+ neurons (*Tg(KalTA4u508;Ihx9Gal4;UAS-GCaMP6s;UAS:ChR2(H134R)-mCherry)* larvae). We used 920 nm two-photon 3D-holographic stimulation to activate a small area ($12 \times 12 \mu\text{m}$; Figures S6A–S6D) within the APN of these larvae, which resulted in a calcium response in these neurons (Figures S6E and S6F). At the same time, we imaged GCaMP at 1,020 nm (Figure 7C, left).³⁸ The optogenetic sessions were followed by a visual stimulation session, in which the fish was presented with prey-like stimuli (Figure 7C,

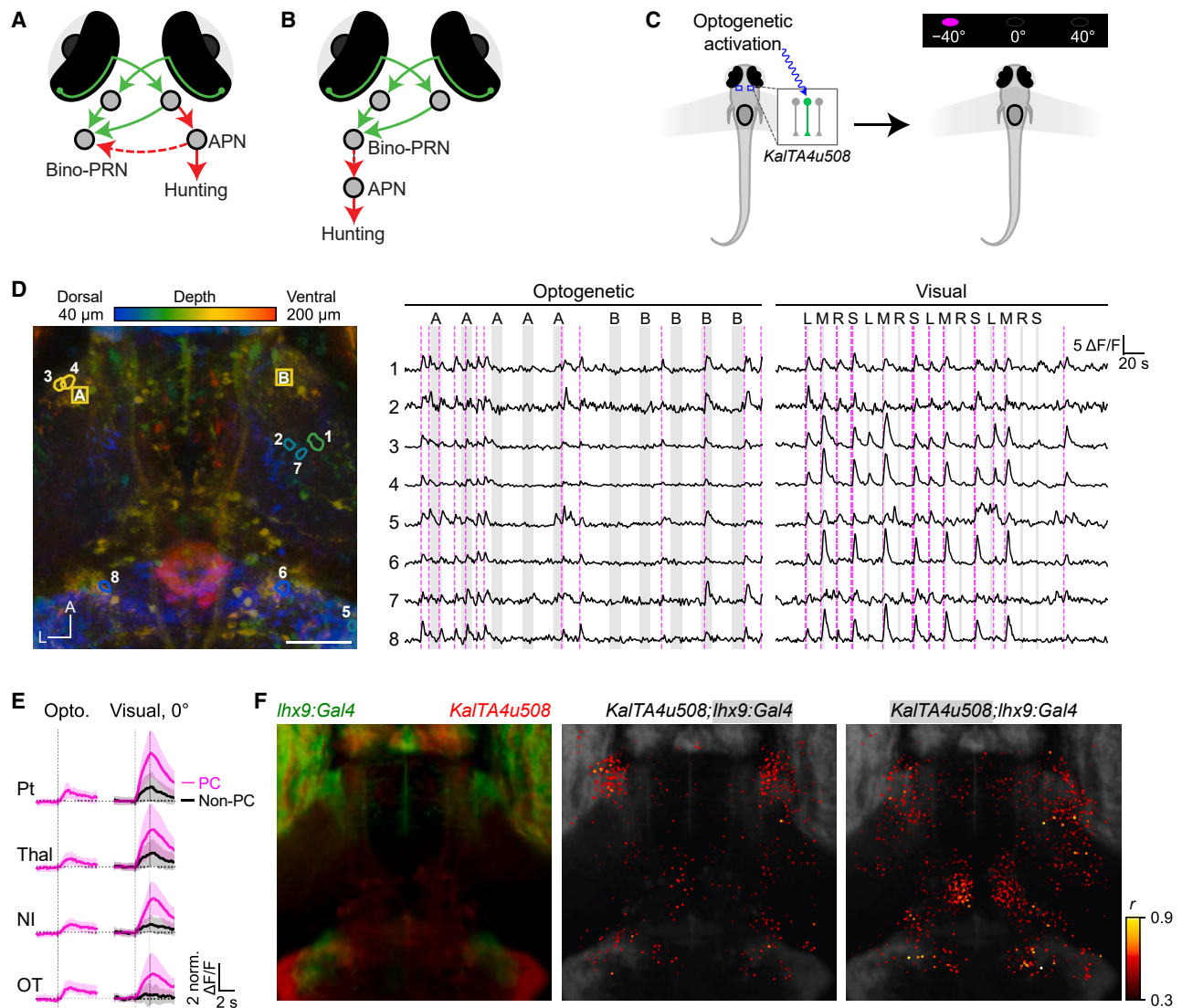


Figure 7. Optogenetically triggered hunting events activate PRNs

(A) Our hypothesis for interactions between command-like APN neurons and bino-PRNs. (B) The alternative hypothesis, in which bino-PRNs activate the APN to cause hunting. (C) Schematic of sequential optogenetic activation and visual stimulation paradigm. *KalTA4u508*+ Pt neurons were holographically activated to trigger PC while performing two-photon calcium imaging. After the opto-imaging session, prey stimuli were presented to record visual responses. (D) Left: example larva expressing GCaMP6s and ChR2 under control of the *KalTA4u508* and *lhx9* lines. Locations of the stimulation volumes (cyan squares, $12 \times 12 \mu\text{m}$) and opto-hunting correlated neurons (circles). Middle: calcium traces of the top five opto-hunting correlated neurons in one fish. Gray shading represents stimulation of the right (A) or left (B) APN. Magenta dashed lines represent hunting onsets. Right: responses to visual prey stimuli in the same five neurons to right, middle, and left circulating prey and sweep prey stimuli (RMLS). (E) Responses of opto-hunting neurons in four regions to opto-hunting events and to visual stimulation. (F) Left: the *lhx9* (green) and *u508* (red) masks. Middle: opto-hunting neurons within the *lhx9* mask. Right: opto-hunting neurons in the *u508* mask. Dorsal view. $n = 5$ fish. Scale bar, $50 \mu\text{m}$. See also Figures S6 and S7.

right). We again freed the tail and kept the eyes in agarose, using eye convergence to define hunting events, and found that these events occurred during the optogenetic experiments, both in and out of phase with the holographic stimulation (Figures 7D, middle, red dashed lines and S7A). These bouts were characterized by eye convergence and J-turn or forward swim tail movements (Figure S7B).

To determine whether these hunting bouts were due to the two-photon scanning activating ChR2, we performed control experiments in larvae with different combinations of the three transgenes. We found that the ChR2 and *KalTA4u508* transgenes were essential for the induction of hunting (Figure S7C). In fish carrying these two transgenes, we did observe hunting bouts in experiments with scanning but no stimulation (Figure S7C, orange

bar), suggesting that these bouts were due to activation of ChR2 by the imaging laser. Indeed, in the absence of two-photon scanning and holographic stimulation, we observed no spontaneous PC bouts (Figure S7C, green). In summary, our holographic stimulation can evoke hunting, but two-photon scanning can also activate the *KalTA4u508* APN neurons to cause some PC bouts during the scanning period.

To identify neurons correlated with opto-hunting behavior, we constructed regressors for PC events and defined neurons with $R^2 \geq 0.3$ as opto-hunting neurons. After activation of a small area within the APN (Figures 7D, left, squares and S7D), we found cell bodies that were correlated with optogenetically evoked eye convergence (Figure 7D, middle). Many of these neurons also responded to our visual prey stimuli (Figure 7D, right), indicating that they are PRNs. Indeed, across all fish, opto-hunting neurons in the four areas where we found them—the Pt, Thal, NI, and OT—responded both to opto-hunting events and to prey stimuli, with a larger response in hunting trials (Figure 7E).

To determine how many of the opto-hunting neurons identified in this experiment were *lhx9*+, we created masks of the brain regions labeled by the two lines (Figure 7F, left; STAR Methods) and found that many of the opto-hunting neurons were located within the *lhx9* mask (Figure 7F, middle), and these neurons were mostly located in the Thal, Pt, and NI (Figure S7E). These results suggest that the *lhx9* neurons are activated by the hunting command, possibly via corollary discharge.

One caveat of the *u508/lhx9* experiments is that some opto-hunting bouts are triggered by the scan, and although the *u508* line was required for opto-hunting (Figure S7C), we did not completely separate the stimulated and responding neuronal populations. To address this concern, we performed similar optogenetic experiments with GCaMP expressed in a separate population of neurons from the *u508* line. We used the larvae with GCaMP expressed panneuronally, and ChR2 only in the *u508* line (*elavl3:Hsa.H2B-GCaMP6s* and *KalTA4u508;UAS:ChR2(H134R)-mCherry*), and we repeated the experiment in Figure 7C with these fish. In this larger population, we found that the opto-hunting neurons shared a similar anatomical pattern with the bino-PRNs identified in Figure 2H (Figure S7F, left). In addition, a subset of these neurons were located within the *lhx9* mask (Figure S7F, right), and approximately half of these opto-hunting neurons responded to visual prey stimuli and were classified as PRNs using the same threshold as Figure 2 (Figure S7G).

Due to space constraints in the holographic optogenetic setup, we were not able to use the eye-blocker to directly determine how many of the opto-hunting neurons were bino-PRNs. However, we show that neurons in the Thal, PT, and NI are activated as a consequence of the hunting motor command, and many of these neurons were within the mask of the *lhx9* line, which labels substantially more bino- than contra- or ipsi-PRNs (Figures 2F–2H). These data suggest that the motor modulation of the bino-PRNs (Figures 5B and 5C) is likely due to a hunting-related corollary discharge from the APN (Figure 7A). It is also possible that the bino-PRNs are command neurons that are activated during scanning, and they activate the APN neurons to cause hunting (Figure 7B). While we cannot eliminate this possibility, we would argue that it is unlikely, given that the

KalTA4u508 line is essential for opto-hunting (Figure S7C). In addition, the *KalTA4u508* line PRNs were predominantly monocular, and those in the *lhx9* line were binocular (Figure 6), and it is unclear how the binocular response could be funneled into a monocular one in Figure 7A.

Coactivation of command-like neurons and *lhx9* evokes PC swims

We next sought to identify the behavioral role of the *lhx9*+ neurons by optogenetically activating them. We used an optic fiber to deliver 470 nm light to the whole brain to activate *KalTA4u508* neurons (Figure S7H). Since we were not using a visual stimulus, we freed the eyes as well as the tail of these larvae. As previously reported, activating *KalTA4u508* neurons drove PC J-turns and forward swims.²⁵ We then performed the same optic fiber activation in larvae with ChR2 expressed in both the *lhx9* and *KalTA4u508* populations, which produced hunting bouts that resembled turns (asymmetrical hunting) or forward swims (symmetrical hunting), as well as non-hunting bouts (other; Figure S7I; Video S2). In the double Gal4 larvae, we found that the optogenetically evoked PC bouts were significantly more symmetrical than for *u508* only (Figure S7J). Larvae bearing only the *lhx9* Gal4 also had higher bout symmetry (Figure S7I). This higher symmetry could be due to a larger proportion of forward swim bouts or a higher degree of symmetry within the forward swim bouts. However, although PRN population within the *lhx9* line consists predominantly of bino-PRNs (Figure 6), the line also contains non-PRNs, and thus we cannot ascribe the symmetrical PC bouts to bino-PRN activation. We then performed two-photon ablations of *lhx9*+ PRNs to determine whether they are mediating the change in swim kinematics. We ablated ~20 *lhx9*+ PRNs in the NI of double Gal4 larvae and then performed the optic fiber experiment. The ablated larvae had more asymmetrical swims than the sham controls (Figure S7K), suggesting that the *lhx9*+ PRNs, which are largely bino-PRNs, contribute to the more symmetrical opto-hunting bouts of the double Gal4 fish, compared with *u508* only, and that bino-PRNs could be involved in the approach phase of PC, when the prey is visible to both eyes.

DISCUSSION

Here, we identify bino-PRNs in the zebrafish visual system by occluding the eyes and volumetrically imaging visual brain areas. We found that these neurons are located primarily in the Pt, Thal, and NI. Notably, our bino-PRN population contained very few neurons in the OT (Figure 2H), although the tectum had the largest share of mono-contra PRNs (Figure 2G). This absence is somewhat surprising given the importance of the tectum in PC behavior.²⁴ It may be that while the tectum is important for some aspects of hunting, the Pt, Thal, and NI are the loci of excitatory binocular interactions. However, the tectum may receive inhibitory interhemispheric input.²⁷

The identification of binocular neurons allows us to analyze their functional properties to evaluate their potential roles in PC. We found that the left eye and right eye receptive fields of the bino-PRNs were offset to varying degrees, with the peak of the distribution at an offset of 30° of the visual field (Figure 4B). In a freely swimming larva, the eyes would converge during

hunting, and in the case of full convergence, the bino-PRNs with a 30° offset would be best activated by prey objects 0.47 mm from the larva. Larvae generally converge their eyes fully near the end of a hunting episode, when they have approached the prey and placed it in the central visual field, and they then typically execute a strike when the prey is 0.5 mm away.¹² The fact that bino-PRNs are tuned for prey objects at the naturalistic strike distance suggests that they could allow the larva to estimate distance to the prey in order to successfully capture it.

Our imaging data also suggested that bino-PRNs were more active in trials where hunting was initiated (Figure 5). Indeed, in trials where hunting was optogenetically triggered, a number of neurons in the Pt, Thal, and NI were activated, and many of these neurons were PRNs (Figure 7). The source of the motor modulation of bino-PRNs also remains an open question.

As the signature of hunting is eye convergence, the motor signal could indicate the degree of convergence, which would be informative for estimating distance to prey. If larvae are using triangulation, where activation of different populations of bino-PRNs corresponds to prey at different distances, it would also be crucial to know how converged the eyes are, as the same set of neurons would be activated by more distant prey when the eyes are not fully converged (Figure 4C, 60° binocular angle). However, convergence angle information would also be informative for disparity-based distance estimation, as it would indicate the distance of the fixation point, which must be combined with disparity information to determine distance to the prey.^{15,39} Therefore, further experiments will be required to elucidate the nature of the binocular computation.

Overall, we have identified a set of bino-PRNs that zebrafish larvae could use to successfully capture prey. Based on their receptive fields, most of these neurons respond optimally to prey at striking distance from a hunting larva, and they may receive a corollary discharge signal indicating eye convergence. These properties could allow the larva to determine whether to execute a strike, with what degree of propulsion, or whether an additional approach swim is required to reach strike range. Additional experiments will be required to determine the exact role of binocular neurons in PC, how these neurons receive input from the ipsilateral eye, the source of the motor signal, and the type of binocular computation. However, our findings lay the foundation for understanding binocular vision in a simple vertebrate visual system.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Julie Semmelhack (jsemmelhack@ust.hk).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Data from one example fish have been deposited at the Harvard Dataverse repository and are publicly available as of the date of publication. The link is listed in the [key resources table](#).
- Custom code used in this study has been deposited at GitHub and is publicly available as of the date of the publication. The link is listed in the [key resources table](#).

- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

ACKNOWLEDGMENTS

Funding was provided by the Hong Kong Research Grant Council to J.L.S. (16103522, 16101221, 16103118, and 26100617) and J.Y.Q. (16102122) and by the Chau Hoi Shuen Foundation. We thank Aristides Arrenberg and members of the Semmelhack lab for feedback on the manuscript.

AUTHOR CONTRIBUTIONS

Conceptualization, G.T.; methodology, G.Y. and Y.H.; software, G.Y. and T.K.C.L.; formal analysis, T.K.C.L.; investigation, G.T., T.K.C.L., and G.Y.; resources, G.Y. and B.K.; writing – original draft, G.T. and J.L.S.; writing – review and editing, all authors; supervision, J.L.S. and J.Y.Q.; and funding acquisition, J.L.S. and J.Y.Q.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cub.2025.05.063>.

Received: November 24, 2024

Revised: March 19, 2025

Accepted: May 29, 2025

Published: June 26, 2025

REFERENCES

- Maier, A., Cox, M.A., Westerberg, J.A., and Dougherty, K. (2022). Binocular Integration in the Primate Primary Visual Cortex. *Annu. Rev.*

- Vis. Sci. 8, 345–360. <https://doi.org/10.1146/annurev-vision-100720-112922>.
2. Cang, J., Fu, J., and Tanabe, S. (2023). Neural circuits for binocular vision: Ocular dominance, interocular matching, and disparity selectivity. *Front. Neural Circuits* 17, 1084027. <https://doi.org/10.3389/fncir.2023.1084027>.
3. Pettigrew, J.D. (1986). The Evolution of Binocular Vision. In *Visual Neuroscience*, J.D. Pettigrew, K.J. Sanderson, and W.R. Levick, eds. (Cambridge University Press), pp. 208–222. <https://doi.org/10.1017/S0317167100027335>.
4. Nityananda, V., Tarawneh, G., Henriksen, S., Umeton, D., Simmons, A., and Read, J.C.A. (2018). A Novel Form of Stereo Vision in the Praying Mantis. *Curr. Biol.* 28, 588–593.e4. <https://doi.org/10.1016/j.cub.2018.01.012>.
5. Feord, R.C., Sumner, M.E., Pusdekar, S., Kalra, L., Gonzalez-Bellido, P.T., and Wardill, T.J. (2020). Cuttlefish use stereopsis to strike at prey. *Sci. Adv.* 6, eaay6036. <https://doi.org/10.1126/sciadv.aay6036>.
6. Allen, K., Gonzalez-Olvera, R., Kumar, M., Feng, T., Pieraut, S., and Hoy, J. L. (2022). A binocular perception deficit characterizes prey pursuit in developing mice. *iScience* 25, 105368. <https://doi.org/10.1016/j.isci.2022.105368>.
7. Johnson, K.P., Fitzpatrick, M.J., Zhao, L., Wang, B., McCracken, S., Williams, P.R., and Kerschensteiner, D. (2021). Cell-type-specific binocular vision guides predation in mice. *Neuron* 109, 1527–1539.e4. <https://doi.org/10.1016/j.neuron.2021.03.010>.
8. Mearns, D.S., Hunt, S.A., Schneider, M.W., Parker, A.V., Stemmer, M., and Baier, H. (2024). Diverse prey capture strategies in teleost larvae. Preprint at bioRxiv. <https://doi.org/10.7554/eLife.98347.1>.
9. Bianco, I.H., Kampff, A.R., and Engert, F. (2011). Prey capture behavior evoked by simple visual stimuli in larval zebrafish. *Front. Syst. Neurosci.* 5, 101. <https://doi.org/10.3389/fnsys.2011.00101>.
10. Patterson, B.W., Abraham, A.O., MacIver, M.A., and McLean, D.L. (2013). Visually guided gradation of prey capture movements in larval zebrafish. *J. Exp. Biol.* 216, 3071–3083. <https://doi.org/10.1242/jeb.087742>.
11. Budick, S.A., and O'Malley, D.M. (2000). Locomotor repertoire of the larval zebrafish: swimming, turning and prey capture. *J. Exp. Biol.* 203, 2565–2579. <https://doi.org/10.1242/jeb.203.17.2565>.
12. Mearns, D.S., Donovan, J.C., Fernandes, A.M., Semmelhack, J.L., and Baier, H. (2020). Deconstructing Hunting Behavior Reveals a Tightly Coupled Stimulus-Response Loop. *Curr. Biol.* 30, 54–69.e9. <https://doi.org/10.1016/j.cub.2019.11.022>.
13. Dowell, C.K., Lau, J.Y.N., Antinucci, P., and Bianco, I.H. (2024). Kinetically distinct saccades are used in a context-dependent manner by larval zebrafish. *Curr. Biol.* 34, 4382–4396.e5. <https://doi.org/10.1016/j.cub.2024.08.008>.
14. Blake, R., and Fox, R. (1973). The psychophysical inquiry into binocular summation. *Percept. Psychophys.* 14, 161–185. <https://doi.org/10.3758/BF03198631>.
15. Nityananda, V., and Read, J.C.A. (2017). Stereopsis in animals: evolution, function and mechanisms. *J. Exp. Biol.* 220, 2502–2512. <https://doi.org/10.1242/jeb.143883>.
16. Cumming, B.G., and DeAngelis, G.C. (2001). The physiology of stereopsis. *Annu. Rev. Neurosci.* 24, 203–238. <https://doi.org/10.1146/annurev.neuro.24.1.203>.
17. Khan, B., Jaesiri, O.M., Lazarte, I.P., Li, Y., Tian, G., Zhao, P., Zhao, Y., Ho, V.D., and Semmelhack, J.L. (2023). Zebrafish larvae use stimulus intensity and contrast to estimate distance to prey. *Curr. Biol.* 33, 3179–3191.e4. <https://doi.org/10.1016/j.cub.2023.06.046>.
18. Bartel, P., Janiak, F.K., Osorio, D., and Baden, T. (2021). Colourfulness as a possible measure of object proximity in the larval zebrafish brain. *Curr. Biol.* 31, R235–R236. <https://doi.org/10.1016/j.cub.2021.01.030>.
19. Khan, B., and Semmelhack, J.L. (2024). The roles of blur and eye convergence in distance estimation in larval zebrafish. *J. Neurogenet.* 38, 195–200. <https://doi.org/10.1080/01677063.2024.2432033>.
20. Kubo, F., Hablitzel, B., Dal Maschio, M., Driever, W., Baier, H., and Arrenberg, A.B. (2014). Functional Architecture of an Optic Flow-Responsive Area that Drives Horizontal Eye Movements in Zebrafish. *Neuron* 81, 1344–1359. <https://doi.org/10.1016/j.neuron.2014.02.043>.
21. Tesmer, A.L., Fields, N.P., and Robles, E. (2022). Input from torus longitudinalis drives binocularity and spatial summation in zebrafish optic tectum. *BMC Biol.* 20, 24. <https://doi.org/10.1186/s12915-021-01222-x>.
22. Fotowat, H., and Engert, F. (2023). Neural circuits underlying habituation of visually evoked escape behaviors in larval zebrafish. *eLife* 12, e82916. <https://doi.org/10.7554/eLife.82916>.
23. Semmelhack, J.L., Donovan, J.C., Thiele, T.R., Kuehn, E., Laurell, E., and Baier, H. (2014). A dedicated visual pathway for prey detection in larval zebrafish. *eLife* 3, e04878. <https://doi.org/10.7554/eLife.04878>.
24. Gahtan, E., Tanger, P., and Baier, H. (2005). Visual Prey Capture in Larval Zebrafish Is Controlled by Identified Reticulospinal Neurons Downstream of the Tectum. *J. Neurosci.* 25, 9294–9303. <https://doi.org/10.1523/JNEUROSCI.2678-05.2005>.
25. Antinucci, P., Folgueira, M., and Bianco, I.H. (2019). Pretectal neurons control hunting behaviour. *eLife* 8, e48114. <https://doi.org/10.7554/eLife.48114>.
26. Henriques, P.M., Rahman, N., Jackson, S.E., and Bianco, I.H. (2019). Nucleus Isthmi Is Required to Sustain Target Pursuit during Visually Guided Prey-Catching. *Curr. Biol.* 29, 1771–1786.e5. <https://doi.org/10.1016/j.cub.2019.04.064>.
27. Gebhardt, C., Auer, T.O., Henriques, P.M., Rajan, G., Duroure, K., Bianco, I.H., and Del Bene, F. (2019). An interhemispheric neural circuit allowing binocular integration in the optic tectum. *Nat. Commun.* 10, 5471. <https://doi.org/10.1038/s41467-019-13484-9>.
28. Förster, D., Helmbrecht, T.O., Mearns, D.S., Jordan, L., Mokayes, N., and Baier, H. (2020). Retinotectal circuitry of larval zebrafish is adapted to detection and pursuit of prey. *eLife* 9, e58596. <https://doi.org/10.7554/eLife.58596>.
29. Muto, A., Lal, P., Ailani, D., Abe, G., Itoh, M., and Kawakami, K. (2017). Activation of the hypothalamic feeding centre upon visual prey detection. *Nat. Commun.* 8, 15029. <https://doi.org/10.1038/ncomms15029>.
30. Zhao, P., Tong, Y., Lazarte, I.P., Khan, B., Tian, G., Chen, K.K.Y., Lam, T.K.C., Hu, Y., and Semmelhack, J.L. (2025). The visuomotor transformations underlying target-directed behavior. *Proc. Natl. Acad. Sci. USA* 122, e2416215122. <https://doi.org/10.1073/pnas.2416215122>.
31. Freeman, J., Vladimirov, N., Kawashima, T., Mu, Y., Sofroniew, N.J., Bennett, D.V., Rosen, J., Yang, C.-T., Looger, L.L., and Ahrens, M.B. (2014). Mapping brain activity at scale with cluster computing. *Nat. Methods* 11, 941–950. <https://doi.org/10.1038/nmeth.3041>.
32. Kunst, M., Laurell, E., Mokayes, N., Kramer, A., Kubo, F., Fernandes, A.M., Förster, D., Dal Maschio, M., and Baier, H. (2019). A Cellular-Resolution Atlas of the Larval Zebrafish Brain. *Neuron* 103, 21–38.e5. <https://doi.org/10.1016/j.neuron.2019.04.034>.
33. Marques, J.C., Li, M., Schaak, D., Robson, D.N., and Li, J.M. (2020). Internal state dynamics shape brainwide activity and foraging behaviour. *Nature* 577, 239–243. <https://doi.org/10.1038/s41586-019-1858-z>.
34. Krauss, A., and Neumeier, C. (2003). Wavelength dependence of the optomotor response in zebrafish (*Danio rerio*). *Vision Res.* 43, 1273–1282. [https://doi.org/10.1016/S0042-6989\(03\)00090-7](https://doi.org/10.1016/S0042-6989(03)00090-7).
35. Baden, T. (2021). Circuit mechanisms for colour vision in zebrafish. *Curr. Biol.* 31, R807–R820. <https://doi.org/10.1016/j.cub.2021.04.053>.
36. Förster, D., Arnold-Ammer, I., Laurell, E., Barker, A.J., Fernandes, A.M., Finger-Baier, K., Filosa, A., Helmbrecht, T.O., Kölsch, Y., Kühn, E., et al. (2017). Genetic targeting and anatomical registration of neuronal populations in the zebrafish brain with a new set of BAC transgenic tools. *Sci. Rep.* 7, 5230. <https://doi.org/10.1038/s41598-017-04657-x>.
37. Fukutomi, M., and Carlson, B.A. (2020). A History of Corollary Discharge: Contributions of Mormyrid Weakly Electric Fish. *Front. Integr. Neurosci.* 14, 42. <https://doi.org/10.3389/fnint.2020.00042>.

38. dal Maschio, M., Donovan, J.C., Helmbrecht, T.O., and Baier, H. (2017). Linking Neurons to Network Function and Behavior by Two-Photon Holographic Optogenetics and Volumetric Imaging. *Neuron* 94, 774–789.e5. <https://doi.org/10.1016/j.neuron.2017.04.034>.
39. Read, J.C.A. (2023). Stereopsis without correspondence. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 378, 20210449. <https://doi.org/10.1098/rstb.2021.0449>.
40. Del Bene, F., Wyart, C., Robles, E., Tran, A., Looger, L., Scott, E.K., Isacoff, E.Y., and Baier, H. (2010). Filtering of visual information in the tectum by an identified neural circuit. *Science* 330, 669–673. <https://doi.org/10.1126/science.1192949>.
41. Thiele, T.R., Donovan, J.C., and Baier, H. (2014). Descending Control of Swim Posture by a Midbrain Nucleus in Zebrafish. *Neuron* 83, 679–691. <https://doi.org/10.1016/j.neuron.2014.04.018>.
42. Lister, J.A., Robertson, C.P., Lepage, T., Johnson, S.L., and Raible, D.W. (1999). nacre encodes a zebrafish microphthalmia-related protein that regulates neural-crest-derived pigment cell fate. *Development* 126, 3757–3767. <https://doi.org/10.1242/dev.126.17.3757>.
43. Khan, B., Lazarte, I.P., Jaesiri, O.M., Zhao, P., and Semmelhack, J.L. (2024). Protocol for using UV stimuli to evoke prey capture strikes in head-fixed zebrafish larvae. *Star Protoc.* 5, 102780. <https://doi.org/10.1016/j.xpro.2023.102780>.
44. Pozzi, P., Maddalena, L., Ceffa, N., Soloviev, O., Vdovin, G., Carroll, E., and Verhaegen, M. (2018). Fast Calculation of Computer Generated Holograms for 3D Photostimulation through Compressive-Sensing Gerchberg-Saxton Algorithm. *Methods Protoc.* 2, 2. <https://doi.org/10.3390/mps2010002>.
45. Pachitariu, M., Stringer, C., Dipoppa, M., Schröder, S., Rossi, L.F., Dalglish, H., Carandini, M., and Harris, K.D. (2017). Suite2p: beyond 10,000 neurons with standard two-photon microscopy. Preprint at bioRxiv. <https://doi.org/10.1101/061507>.
46. Giovannucci, A., Friedrich, J., Gunn, P., Kalfon, J., Brown, B.L., Koay, S.A., Taxis, J., Najafi, F., Gauthier, J.L., Zhou, P., et al. (2019). CalmAn an open source tool for scalable calcium imaging data analysis. *eLife* 8, e38173. <https://doi.org/10.7554/eLife.38173>.
47. Avants, B.B., Tustison, N.J., Song, G., Cook, P.A., Klein, A., and Gee, J.C. (2011). A reproducible evaluation of ANTs similarity metric performance in brain image registration. *Neuroimage* 54, 2033–2044. <https://doi.org/10.1016/j.neuroimage.2010.09.025>.
48. Shainer, I., Kuehn, E., Laurell, E., Al Kassab, M., Mokayes, N., Sherman, S., Larsch, J., Kunst, M., and Baier, H. (2023). A single-cell resolution gene expression atlas of the larval zebrafish brain. *Sci. Adv.* 9, eade9909. <https://doi.org/10.1126/sciadv.ade9909>.
49. Triplett, M.A., Pujic, Z., Sun, B., Avitan, L., and Goodhill, G.J. (2020). Model-based decoupling of evoked and spontaneous neural activity in calcium imaging data. *PLOS Comput. Biol.* 16, e1008330. <https://doi.org/10.1371/journal.pcbi.1008330>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Invitrogen Ultrapure Low Melting Point Agarose	ThermoFisher Scientific	Cat#16520050
Hepes	Sigma-Aldrich	H4034
Sodium chloride	VMR Chemicals	Cat#27810.262
Potassium chloride	VMR Chemicals	Cat#26764.298
Magnesium sulphate heptahydrate	VMR Chemicals	Cat#25165.260
Calcium nitrate tetrahydrate	Sigma-Aldrich	Cat#237124
Methylene blue	Sigma-Aldrich	M9140
Deposited data		
Data from one example fish	This paper	https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/MRXESB
Experimental models: Organisms/strains		
<i>Tg(elavl3:Hsa.H2b-GCaMP6s)jf5Tg</i>	Freeman et al. ³¹	N/A
<i>Tg(atox7:GAL4-VP16)s1992tTg</i>	Del Bene et al. ⁴⁰	N/A
<i>Tg(UAS:GCaMP6s)mpn101</i>	Thiele et al. ⁴¹	N/A
<i>Tg(KalTA4u508)u508Tg</i>	Antinucci et al. ²³	N/A
<i>Tg(lhx9:Gal4VP16)mpn203</i>	Förster et al. ³⁶	N/A
<i>Tg(UAS:Chr2(H134R)-mCherry)mpn134</i>	dai Maschio et al. ³⁸	N/A
Software and algorithms		
Custom imaging analysis code	This paper	https://doi.org/10.5281/zenodo.15483912
Other		
Eye blocker design files	This paper	https://github.com/SemmelhackLab/eye-blocker-model

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

All procedures conformed to the guidelines by the animal ethics committee of HKUST. Animals were kept under a standard 14:10 light cycle at 28°C. Zebrafish larvae carrying mutations in the *mitfa* allele (*nacre*) were used for all experiments.⁴² Larvae were kept in Danieau's solution, and fed with paramecia (*Paramecium multimicronucleatum*, Carolina Biological Supply Company) on 5 to 7 days post fertilization (dpf).

The following transgenic lines were used for this study: *Tg(atox7:GAL4-VP16)s1992tTg*,⁴⁰ *Tg(UAS:GCaMP6s)mpn101*,⁴¹ *Tg(elavl3:Hsa.H2b-GCaMP6s)jf5Tg*,³¹ *Tg(KalTA4u508)u508Tg*,²⁵ *Tg(lhx9:Gal4VP16)mpn203*,³⁶ *Tg(UAS:Chr2(H134R)-mCherry)mpn134*.³⁸ For imaging of RGC axons and pan-neuronal imaging, *Tg(atox7:Gal4VP16;UAS:GCaMP6s)* and *Tg(elavl3:Hsa.H2b-GCaMP6s)* fish were used, respectively. *Tg(lhx9:Gal4VP16;UAS:GCaMP6s)* and *Tg(KalTA4u508;UAS:GCaMP6s)* larvae were screened for binocular neurons. For two-photon optogenetic stimulation with calcium imaging experiments, *Tg(KalTA4u508;lhx9:Gal4VP16;UAS:GCaMP6s;UAS:Chr2(H134R)-mCherry)*, *Tg(KalTA4u508;UAS:GCaMP6s;UAS:Chr2(H134R)-mCherry)*, *Tg(lhx9:Gal4VP16;UAS:GCaMP6s;UAS:Chr2(H134R)-mCherry)* and *Tg(elavl3:Hsa.H2b-GCaMP6s; KalTA4u508;UAS:Chr2(H134R)-mCherry)* fish were used. For optic fiber stimulation experiments, *Tg(KalTA4u508;lhx9:Gal4VP16;UAS:Chr2(H134R)-mCherry)*, *Tg(KalTA4u508; UAS:Chr2(H134R)-mCherry)* and *Tg(lhx9:Gal4VP16;UAS:Chr2(H134R)-mCherry)* fish were used.

METHOD DETAILS

Zebrafish preparation

6–8 dpf larvae were embedded in 2% low melt agarose (Invitrogen) on 100 mm square (Thermo Scientific (cat# 109-17)) or 90 mm round (Thermo Scientific (cat#101VR20)) plastic petri dishes. For all experiments involving visual stimulation, the agarose surrounding the tail and in front of the head was removed by surgical blade (Paragon p303) to allow for tail and fin movements while eyes were

kept in the agarose. For the optic fiber stimulation experiments (Figure S9) only, both eyes and tail were freed. Fish were allowed to recover overnight prior to the experiment.

Eye-blocker design

Files describing the design of the eye-blocker are deposited in github and available via the link in the [key resources table](#). The components of the eye-blocker were fabricated using a 3D printer with black PLA plastic as the material. The eye-blocker was assembled from two arm parts and a body part. During imaging experiments, the tips of the eye-blocker arms were placed between the objective and the bottom of the fish arena. The height of the tips was chosen to minimize the gaps at the top and the bottom without hindering the vertical movements of the objective. The body of the eye-blocker was attached to a 3-axes stage via an optical post. By controlling the movements of the 3-axes stage along the left-right axis of the fish, the arms of the eye-blocker could be moved to the position required for each ocular condition. The movements of the eye-blocker were visually guided by the live preview from the high-speed camera.

Lensexotomy

At 6–7 dpf, larvae were embedded in 2% low melting agarose, and then the prep was immersed in 0.02% tricaine (MS-222 Sigma-Aldrich) to anaesthetize the larvae throughout the surgery. Lensexotomies were performed with needle (Terumo 0.40 x 13mm). An incision was made in the transparent cuticle of the temporal side of the eye and lens was scouped out gently. Animals were then released from the agarose, allowed to recover overnight, and fed paramecia. The delensed larvae were re-embedded the next day (7–8 dpf), and tested the day after (8–9 dpf).

Visual stimulation

We adopted the spatial visual stimulator module (DLP® Light Crafter™ 4500 modified by EKB Technologies Ltd) to present stimuli as previously described.⁴³ 385 nm “UV” LED (SOLIS-385C Thorlabs) and 590 nm “Red” LED (M590L4 Thorlabs) were used as the light sources. The Light Crafter (LCr) back projected through one wall of the petri dish to the 50 mm wide x 5 mm height Teflon fabric screen at 13.5 cm away. The screen was positioned inside the petri dish, underwater, and 1 cm in front of the fish head.

Design of visual stimuli

Visual stimuli were created using PsychoPy. The prey-like circulating stimuli were UV dots, sized $4^\circ \times 2^\circ$, moving in an elliptical path of $8^\circ \times 4^\circ$ at a speed of 2 Hz for 2 seconds on a dark background. The long axis of the dots was aligned to the movement direction. The power of the UV light was 2.44 nW/mm^2 , and the background was black (measured as 0.016 nW/mm^2 at 385 nm). In the eye-blocking experiment (Figures 1, 2, 3, and 4), the stimuli were presented at different positions, ranging from -50° (left) to 50° (right) in 10° steps. Stimuli on the left side moved counterclockwise, while those on the right moved clockwise. For the center azimuth (0°), we presented both counterclockwise and clockwise directions (indicated as -0° and $+0^\circ$), resulting in 12 distinct stimuli. We randomly presented these 12 stimuli four times in each imaging session, with a 13 s interstimulus interval (ISI). We conducted two sessions for each eye condition, with at least a 5-minute pause between sessions, and followed a palindromic sequence for the eye conditions. The first stimulus in all experiments was shown 30 seconds after starting the two-photon scanning.

In the stimulus specificity experiment (Figure S4), we conducted one imaging session each for left-eyed, right-eyed, and binocular conditions. The stimuli were the identical to those used in the eye-blocking experiment, except the background was red (1.78 nW/mm^2 at 590 nm). After the sessions with the circulating dots, we presented a battery of eight additional types of stimuli under the binocular condition: darkening and brightening of the screen, leftwards and rightwards sweeping dots, leftwards and rightwards moving gratings, and slow and fast looming disks. The darkening and brightening stimuli changed the screen from red to black, or the reverse, over 10 seconds, with an ISI of 10 s. The sweeping dots have a size of $4^\circ \times 2^\circ$ and moved horizontally from -60° to 60° , or the reverse, over 2 s. The grating pattern was a sinusoid alternating between red (1.78 nW/mm^2 at 590 nm) and black with a cycle of 15° moving horizontally at $20^\circ/\text{s}$ for 10 s. It remained visible but stationary during the 10-second ISI. The looming stimuli were dark circles that expanded on a red background (1.78 nW/mm^2 at 590 nm) from the screen's center at slow (l/v ratio = 0.25) and fast (l/v ratio = 0.125) speeds, with a 30-second interval between their initiations.

In the optogenetics imaging experiment (Figure 6), we conducted a visual stimulation session after the optogenetic stimulation sessions. We used a sweeping stimulus in addition to the -40° , 0° , and 40° circulating stimuli. The sweeping stimulus was a $4^\circ \times 2^\circ$ UV dot moving horizontally from -60° to 60° over 2 s. The stimulus sequence of (-40° , 0° , 40° , sweep) was presented four times with an ISI of 13 s.

Behavioral recording

Behavior was recorded at 200 Hz using a high-speed camera (PhotonFocus MV1-D1312-160-CL) at a resolution of 544×560 pixels, covering the entire fish and a small area in front to track the eye-blocker. Illumination was provided by an 850 nm infrared LED array beneath the fish arena, and an infrared band-pass filter on the camera lens blocked light from the two-photon laser.

Calcium imaging

For the eye-blocker experiments, in-vivo calcium imaging was performed using a two-photon microscope (Nikon A1R MP+) equipped with a $25\times$ water-immersion objective (Nikon CFI75 Apochromat 25XC W 1300, 1.1 numerical aperture). The excitation laser was

calibrated to a wavelength of 1020 nm. An electrically tunable lens (Optotune EL-16-40-TC) was mounted behind the objective to enable multi-plane functional imaging. The imaged volume contains 9 planes with an average spacing of 15 μm between consecutive planes. Each plane was imaged at a resolution of 512×512 pixels (0.99 $\mu\text{m}/\text{pixel}$) at 3 Hz. Anatomical z-stacks were collected after the imaging sessions.

For the optogenetics experiments as in Figure 5, a home-built two-photon light scanning microscope controlled by a C# based custom software is employed. The laser source is a Ti:Sapphire femtosecond laser (Chameleon Ultra II, Coherent) operating at 1020 nm wavelength for calcium imaging to reduce the crosstalk between GCaMP6s and ChR2. The laser beam first passes through the half-wave plate to adjust polarization direction. The laser beam with 2.2 mm in diameter (propagate a large distance from the laser source and expanded from 1.2 mm) is expanded by a factor of $\times 7.14$ through a 4f system (AC254-035-B-ML F = 35 mm, Thorlabs and AC254-250-B-ML F = 250 mm, Thorlabs) to full fill the ETL (EL-16-40-TC-NIR, Optotune AG), which is used for multi-plane functional imaging. Then a 4f system (AC254-200-AB-ML F = 200 mm, Thorlabs, and AC254-050-AB-ML F = 50 mm, Thorlabs) reduces the beam diameter to 4 mm to avoid overfilling the scanner mirror. A resonant scanner (CRS 8 kHz, Cambridge technology) for high-speed scanning is conjugated with a set of XY galvo scanners (6215H, Cambridge technology) by a 1:1 relay (AC254-100-AB-ML F = 100 mm, Thorlabs). Finally, two lenses (Scan lens SL50-CLS2 F = 50 mm, Thorlabs, and Tube lens TTL200MP F = 200 mm, Thorlabs) are used to correct the aberration and expand the laser beam to fill the objective back pupil (objective lens (20x, N.A. 1.0, XLUMPLFLN20XW, Olympus Corporation)). The ETL, resonant scanner, XY galvo scanners, and objective pupil plane are all conjugated. In the fluorescence collection path, a dichroic mirror (FF705-Di01-25 $\times$ 36, Semrock) is set between the tube lens and objective to reflect the fluorescence light, which then passes through two relay lenses (AC254-150-AB-ML F = 150 mm, Thorlabs and AC254-040-AB-ML F = 40 mm, Thorlabs) into the PMT (H11461-03, Hamamatsu). Two filters (FF03-525/50-25, Semrock, and FF01-600/52-25, Semrock) are installed on a flipper which can switch the filter to collect different fluorescence signals. Voltage signals are converted from PMT output current using a high-speed current amplifier (DHPCA-100, Femto) and digitized using an oscilloscope device (PCIe-8512H, ART technology).

Two-photon optogenetic stimulation

The laser source is a high-power ultrafast fiber laser (FemtoFiber ultra 920, TOPTICA Photonics AG). The laser beam first passes through the half-wave plate to adjust polarization direction. And then pass through a set of galvo scanner (6215H, Cambridge Technology). Then the light beam is expanded by two lenses (AC254-50-AB-ML F = 50 mm, Thorlabs, and AC254-250-AB-ML F = 250 mm, Thorlabs) to fill the spatial light modulator (HSP1K-500-1200-PC8, Meadowlark Optics). Then the SLM is conjugated to the back focal plane of objective by a demagnify relay (AC508-250-AB-ML F = 250 mm, Thorlabs, and AC254-150-AB-ML F = 150 mm, Thorlabs). The photon stimulation light combines with imaging light by a polarized beam splitter (CCM1-PBS252/M; Thorlabs) between tube lens and dichroic mirror.

The hologram is calculated by compressive sensing weighted Gerchberg–Saxton algorithm⁴⁴ and loaded on SLM which controlled by a custom software on MATLAB. To compress the axial profile, a large beam diameter is applied to achieve an effective excitation NA (~ 0.5). Then by generating a spot array on each target, the target neuron can be fulfilled with a square of 10–12 μm diameter and with an average power density of 0.15–0.3 $\text{mW}/\mu\text{m}^2$.

The spatial coordination between imaging light and photo stimulation light is calibrated before each optogenetics experiment. Two dark points are generated by photon bleaching a fluorescent microscope slide (FSK3, Thorlabs) for both imaging light and photo stimulation light. After calculating the displacement between them, a coordination shift is added to the hologram calculation.

During the two-photon optogenetics experiment, for each zebrafish larva, 6–8 sessions are conducted depending on the specific expression. For each session, a single pretectal *KalTA4u508* neuron is randomly selected according to mCherry expression and activated. Each session contains 5 trials, whereby each trial entails a 10-second duration of neuronal activation followed by a 30-second intertrial interval.

Single-photon optogenetic stimulation

Zebrafish larvae were placed in the petri dish and fixed with agarose. The eye and tail are freed by carefully cutting the agarose. An optical fiber, 200 μm in diameter, is held 5 mm above the zebrafish larvae and guided the light from a blue light fiber-coupled LED (M470F4, Thorlabs). The irradiance at zebrafish larvae is 12–20 mW/cm^2 and adjusted by pulse-width modulation. Each trial contains a 2 s stimulation period and there is a 15 s interval between each trial. In total 60 trials are conducted for each fish. The whole petri dish is illuminated by an 850 nm LED array. Behavior is recorded at 100 Hz frame rate and analysed by custom Python software. The ChR2 expression is recorded after the experiment by two-photon imaging as a reference. See also Video S2.

Behavioral tracking

We tracked behavior using ztrack, a Python-based custom software. For eye tracking, we applied image thresholding around the fish's head, extracting the three largest contours from the image. The closest pair was identified as eyes, and the third as the swim bladder. Using the cv2.fitEllipse function from OpenCV, we fitted ellipses to these contours to find the centers and orientations of the eyes and swim bladder. We distinguished left from right eye based on the direction of the cross product between the vectors from the swim bladder to each eye. The fish's midline was defined as the line from the swim bladder's center to the midpoint between the eyes. The eye angles were computed as the angles between the major axes of the eyes and the midline. To track the tail, we used an iterative algorithm that traced 11 points along the tail, beginning at the most posterior point of the swim bladder. At each step, the

algorithm identifies the next point on the tail by finding the brightest pixel among locations that were equidistant from the current point.

Behavioral data analysis

Eye convergence events were detected as peaks in the convolution of gaussian filtered ($\sigma = 0.04$ s) vergence angle (L + R eye angle) with a step function (width 1 s), similar to,¹³ and manually corrected. Tail bouts were detected by thresholding the rolling standard deviation of tail angle over a 50 ms window. The asymmetry of each bout was calculated by first integrating the curvature over the last quarter of the tail and then determining the fraction of time points with curvature integrals that matched the sign of the bout's mean curvature.

Processing of calcium imaging data

For each imaging plane, the non-rigid motion correction was performed using suite2p.⁴⁵ In Figure 1E, pixel intensities were preprocessed by z-score normalization across frames prior to analysis. In Figure 1F, AF7 activity was calculated as the mean intensity of pixels within the manually annotated AF7 mask, followed by z-score normalization across time. The tuning curve in Figure 1G was derived from the peak normalized AF7 activity within the stimulus interval.

For neuron-based analyses, the locations of regions of interest (ROIs) and their $\Delta F/F$ were extracted using the CNMF algorithm from the CalmAn package.⁴⁶ The 10th percentile of a rolling 3-minute time window was used as the baseline for calculating $\Delta F/F$. ROIs exhibiting a signal-to-noise ratio (SNR), as computed by CalmAn, below 2.5 were excluded from further analysis. The $\Delta F/F$ for each ROI was normalized by subtracting the mean and then dividing by the standard deviation obtained across all imaging sessions, followed by smoothing with a Gaussian filter ($\sigma = 0.33$ s).

Image registration

Image registration was performed using the antsRegistration function from Advanced Normalization Tools (ANTs)⁴⁷ using parameters described.⁴⁸ Anatomical z-stacks were registered to the templates provided by the mapZebbrain atlas.³²

Assignment of neurons to anatomical regions

For each region of interest (ROI), we calculated its x and y coordinates based on the center of mass of its spatial component. To determine the z coordinates, we identified the depth of each imaging plane by finding the z-stack layer that maximizes the Pearson correlation with the average fluorescence image of that plane. We then converted the 3D coordinates from the z-stack space to the atlas space using the function antsApplyTransformsToPoints. Lastly, we identified the anatomical region of each ROI by referencing annotated volume of the atlas at the transformed 3D coordinates.

Spatial p-value analysis

After aligning all neurons to the mapZebbrain template,² we calculated the mean distance of each ocular condition-PRN to its nearest neighbors exhibiting the same response property across all other fish. This distance was then compared to a null model generated by random sampling, to derive a spatial *P* value. A spatial *P* value of less than 0.025 indicates that the colocalization of the examined neuron with its neighbors is unlikely to be random; thus it is retained as a spatially consistent neuron. For all maps showing the anatomical locations of functionally classified neurons from multiple fish, we retain only the neurons that demonstrate spatial co-localization across fish, as determined through the spatial *p*-value analysis described.³³

Regression analysis

We generated stimulus regressors by convolving boxcar functions indicating stimulus presence with a calcium impulse response function (CIRF), which was modeled as a difference-of-exponentials.⁴⁹ We conducted multiple linear regression analyses using sklearn.linear_model.LinearRegression. In the analysis in Figure 2, we used 12 stimulus regressors ($-50^\circ, \dots, -0^\circ, +0^\circ, \dots, +50^\circ$) to model the $\Delta F/F$ for each ROI. Separate regression models were used for different ocular conditions, yielding one R^2 value per condition. For the analysis in Figure 3, we concatenated the $\Delta F/F$ for each neuron across all ocular conditions. This concatenated data was then modeled as a weighted sum of 38 regressors, composed of 12 stimulus regressors for each of the three ocular conditions (left-eye (LE), right-eye (RE), and binocular) plus 2 motor regressors. For the analysis in Figure S4C, we used 8 stimulus regressors corresponding to the 8 stimulus types. An intercept term was included in all regression models.

Ocular specificity assignment

To categorize neurons into one of four ocular types in Figure 2, we first compute the R^2 values for ipsilateral and contralateral conditions. For neurons in the left hemisphere, R^2_{ipsi} refers to R^2_{LE} , and R^2_{contra} refers to R^2_{RE} . For neurons on the right, these assignments are reversed. Neurons are then classified as follows: binocular neurons: both R^2_{ipsi} and $R^2_{\text{contra}} \geq 0.3$; mono-ipsi neurons: $R^2_{\text{ipsi}} \geq 0.3$ and $R^2_{\text{contra}} < 0.3$; mono-contra neurons: $R^2_{\text{contra}} \geq 0.3$ and $R^2_{\text{ipsi}} < 0.3$. Neurons that do not fit any of these criteria are classified as "Other".

Hierarchical clustering

In [Figure 3](#), we clustered neurons by their responses to the eight stimulus types using affinity propagation followed by hierarchical clustering, as described.²⁸

Analysis of imaging data from optogenetics experiments

For the analysis in [Figure 7F](#), neurons were categorized as *lhx9:Gal4* or *Ka/TA4u508* depending on which corresponding z-scored fluorescence z-stack exhibited a higher z-scored fluorescence intensity at the neuron's location.

QUANTIFICATION AND STATISTICAL ANALYSIS

The statistical tests used can be found in the figure legends.